

2024



crypto.com

Crypto.com Builds AI Engine to Accelerate Video and Image Production by Using Generative AI on AWS

Crypto.com uses a fully managed infrastructure for generative AI on AWS to build a scalable visual content engine, delivering high-quality images and videos faster for multiple use cases to enhance customer engagement and expand.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Outcome](#) | [AWS Services Used](#)

50%

reduction in large ML model loading time



Hi, I can connect you with an AWS representative or answer questions you have on AWS.



40%

decrease in deployment and scale out time



Need more info? Highlight any text to get an explanation generated with AWS generative AI.



to generate 30 high-quality video frames



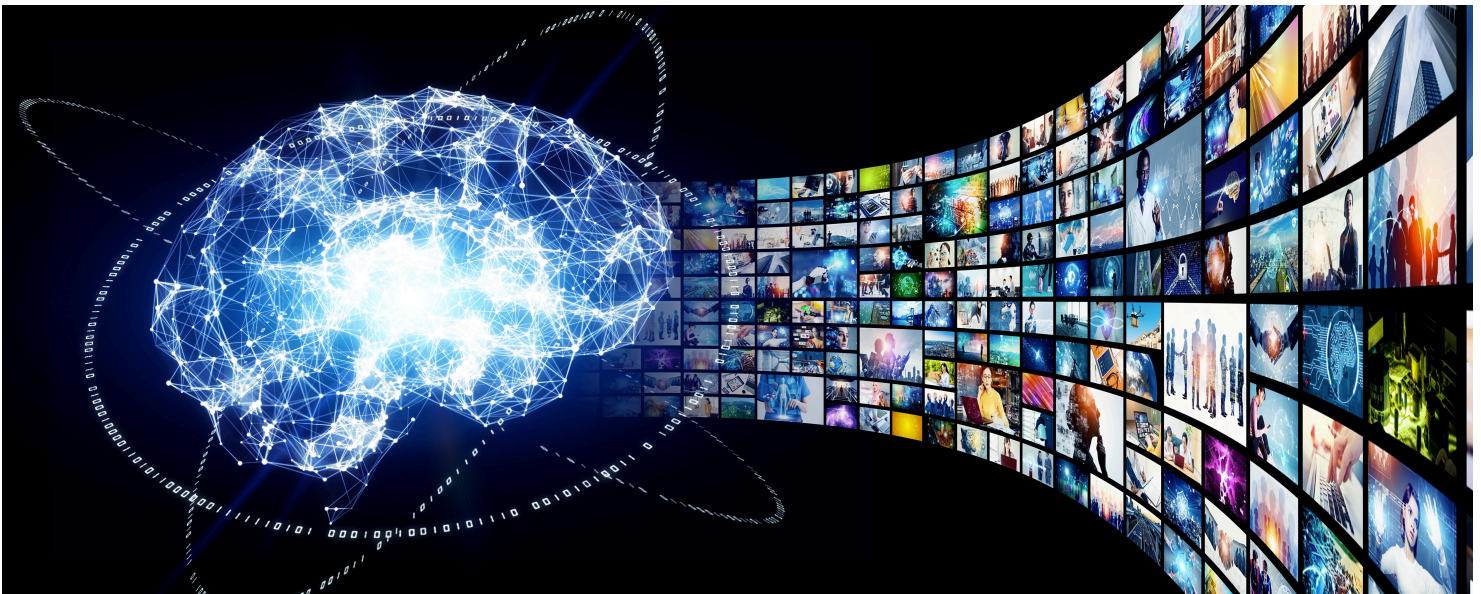
2

1 minute

to create four high-resolution images

Overview

[Crypto.com](#) is a crypto exchange and comprehensive trading platform serving 100 million users in 90 countries. The company faced challenges productionizing its infrastructure for AI-driven visual content generation prototypes, slowing deployment and limiting scalability. By building its visual content generation engine using a fully managed infrastructure for generative AI on Amazon Web Services (AWS), Crypto.com automated workflows, reduced large ML model loading times by 50 percent, and accelerated video and image production, cutting the time to launch new features by 50 percent.



Opportunity | Developing a Cutting Edge AI-Driven Engine for Visual Content Creation

[Crypto.com](#), a global leader in cryptocurrency services with over 100 million users, offers a wide range of digital financial services to accelerate the world's transition to cryptocurrency. To stay competitive in the evolving web3 landscape, the company sought to leverage generative AI to enhance its platform and improve internal productivity and customer experience. Already utilizing AWS for various infrastructure needs, including [Amazon Bedrock](#) for text-based AI applications, Crypto.com aimed to expand into visual content generation—creating AI-driven generation of images and videos to boost engagement and efficiency.

The business successfully developed cutting-edge AI-driven engine prototypes for generating images and videos on [Amazon Elastic Compute Cloud](#) (Amazon EC2). However, the company faced several challenges: manual setup of development environments on Amazon EC2 slowed deployment, its infrastructure wasn't optimized for large AI models, and content moderation was needed to safeguard brand quality.

Furthermore, manual operations like setting up operating system dependencies and managing secure remote connections for different projects were time-consuming, hampering the team's ability to innovate quickly. Additionally, the lack of an optimized infrastructure limited the efficiency and scalability of building and deploying larger generative AI models for its platform.



To overcome these hurdles, Crypto.com required a fully managed solution that would automate development processes, scale efficiently, and streamline the integration of open-source AI models for rapid testing. “Our goal is to optimize our AI tech stack for rapid development and deployment, swiftly integrate the latest AI models, and ensure seamless content moderation to deliver innovative use cases that elevate customer engagement and value,” Sunny Fok, head of AI Innovation Technology at Crypto.com.

Solution | Automating Generative AI Workflows and Enhancing Infrastructure Efficiency

Crypto.com collaborated with AWS to enhance its platform capabilities for visual content generation and streamline its AI development process. By using [Amazon SageMaker](#) and its suite of tools—such as [Amazon SageMaker notebooks](#) in JupyterLab for exploring data and test large image and video models, and [Amazon SageMaker Asynchronous Inference](#) endpoints for serving requests—Crypto.com automated workflows and built advanced pipelines for image and video production. Additionally, by standardizing applications into Docker container images on [Amazon Elastic Container Registry](#) (Amazon ECR) and deploying seamlessly on [Amazon EC2 G5 instances](#) via Amazon SageMaker Endpoint, the company reduced manual processes, model loading times, and adoption time for the latest open-source plugins.

To further accelerate its visual content generation engine development, Crypto.com implemented [Amazon FSx for Lustre](#), creating a fully managed shared storage environment for developing, training, and deploying its large image and video models. Using Amazon SageMaker's built-in MLOps capabilities, the company efficiently trained its model with [Amazon SageMaker HyperPod](#). It also utilized containerized workloads and automated instance provisioning to streamline infrastructure management. Throughout the video training journey, scientists from the AWS Generative AI Innovation Center (GenAIIC) have provided continuous support to the AI & Innovation team.

Crypto.com integrated open-source models from platforms like Hugging Face and GitHub with Amazon SageMaker, facilitating rapid testing and fine-tuning of AI models for image generation, video analytics, and content moderation. “Amazon SageMaker has facilitated us to build a visual generation platform to support different scenarios and use cases for internal and external demands while maintaining a high-performance infrastructure to support our evolving AI initiatives.” says Vincent Leung, AIOps lead at Crypto.com.



Outcome | Accelerating AI-Driven Video and Image Production

Crypto.com's visual content generation engine, built on AWS, has significantly enhanced the company's ability to deliver high-quality video and image content at a faster pace. Crypto.com can now generate 30 high-quality video frames in just two minutes and create four high-resolution images in one minute, cutting large image model loading times by 50 percent compared to previous temporary Amazon EC2-based solutions. This speed improvement allows the development team to react to changes more efficiently, reducing new features launch time by approximately 50 percent.

In addition to faster content generation, Crypto.com implemented a more efficient reusable tagging system for video assets, allowing for faster search and filtering of video libraries, which has greatly improved the team's workflow. The company also integrated a robust moderation system for its AI-generated content, ensuring all media adheres to brand guidelines and maintains high-quality standards. This automation of content moderation has helped Crypto.com scale while preserving its brand integrity.

By utilizing Amazon SageMaker's built-in MLOps capabilities and containerized deployments, Crypto.com achieved a 40 percent reduction in deployment and scale out time, allowing for more frequent updates and agile product development.

Crypto.com continues to explore generative AI solutions on AWS to enhance its platform further. The company plans to expand AI-driven features such as a corporate assistant using retrieval-augmented generation (RAG) techniques and a fully integrated video generation platform for broader teams. Fok says, "By adopting the latest managed service for generative AI on AWS, we're strengthening Crypto.com's position as a leader in the cryptocurrency industry while driving greater efficiency and innovation across our products."

About Crypto.com

Crypto.com is on a mission to accelerate the world's transition to cryptocurrency. With approximately 100 million users across 90 countries, the Singapore-based company offers a trading platform, derivative exchanges, and more. Crypto.com also partners with merchants and payment gateways to support the development of a global crypto network.

AWS Services Used

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The next generation of Amazon SageMaker is the center for all your data, analytics, and AI

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Amazon FSx for Lustre

Fully managed shared storage built on the world's most popular high-performance file system



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Amazon SageMaker HyperPod

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